

# Part II: Brief Approximation Theory

RTG Summer School

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# **Universal Approximation Theorem (UAT)**

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# Historical Context

- **1943:** McCulloch-Pitts neuron (first appearance of artificial neural network, no learning)
- **1958:** Rosenblatt's Perceptron (learns linear boundaries)
- **1969:** Minsky & Papert, *Perceptrons* book (proves single-layer networks cannot learn XOR)

$$f(x) = \sigma(w \cdot x + b)$$

## The Critical Question

Can adding a hidden layer resolve this fundamental limitation?

# The Breakthrough (1989)

Independent simultaneous proofs:

- **George Cybenko:** Sigmoidal activation  $\implies$  universal approximator
- **Hornik, Stinchcombe & White:** Multi-layer feedforward networks
- **Ken-Ichi Funahashi:** Independent proof

## Short Answer

A single hidden layer with any **non-linear activation** suffices to approximate **any** continuous function to arbitrary precision.

## Definition (Space of Continuous Functions)

Let  $K \subset \mathbb{R}^n$  be compact.  $C(K)$  is the space of all continuous real-valued functions on  $K$ , with the **supremum norm**:

$$\|f\|_{\infty} = \sup_{x \in K} |f(x)|$$

Under this norm,  $C(K)$  is a **Banach space**.

**Approximation:** Find network  $N(x)$  such that  $\|f - N\|_{\infty} < \epsilon$ . This guarantees worst-case error bounded across the entire domain.

## Definition ( $L^p$ Spaces)

For a measure space  $(X, \mathcal{M}, \mu)$  and  $1 \leq p < \infty$ :

$$\|f\|_p = \left( \int_X |f(x)|^p d\mu(x) \right)^{1/p} < \infty$$

Functions equal  $\mu$ -almost everywhere are identified.

Machine learning optimizes *expected* loss over probability distributions — understanding  $L^p$  approximation is essential.

# Riesz Representation Theorem

## Theorem (Riesz-Markov-Kakutani)

Let  $K$  be a compact Hausdorff space. For every bounded linear functional  $L$  on  $C(K)$ , there exists a **unique** regular finite signed Borel measure  $\mu$  such that:

$$L(f) = \int_K f(x) d\mu(x)$$

and  $\|L\| = |\mu|(K)$ .

It connects abstract functionals to concrete measures — the bridge between Hahn-Banach separation and the UAT proof.

# Hahn-Banach Separation Theorem

## Theorem (Hahn-Banach — Subspace Corollary)

*Let  $X$  be a real Banach space and  $S \subset X$  a linear subspace. If  $\overline{S} \neq X$ , there exists a non-zero bounded linear functional  $L \in X^*$  such that:*

$$L(x) = 0 \quad \text{for all } x \in S$$



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## Proof Strategy

Assume networks are NOT dense  $\rightarrow$  Hahn-Banach gives annihilating functional  $\rightarrow$  Riesz gives measure  $\mu \rightarrow$  prove  $\mu = 0 \rightarrow$  contradiction.

## Definition (Feedforward Neural Network Function)

$$G(x) = \sum_{i=1}^N \alpha_i \sigma(w_i^T x + b_i)$$

$x \in \mathbb{R}^n, w_i \in \mathbb{R}^n, b_i \in \mathbb{R}, \alpha_i \in \mathbb{R}.$

$S_\sigma =$  linear subspace of all such functions in  $C(K).$

### Definition (Sigmoidal Function)

$\sigma : \mathbb{R} \rightarrow \mathbb{R}$  is **sigmoidal** if:

$$\lim_{t \rightarrow \infty} \sigma(t) = 1 \quad \text{and} \quad \lim_{t \rightarrow -\infty} \sigma(t) = 0$$

### Definition (Discriminatory Function)

$\sigma$  is **discriminatory** if for any finite signed Borel measure  $\mu$ :

$$\int_K \sigma(w^T x + b) d\mu(x) = 0 \quad \forall w, b \implies \mu = 0$$

# Cybenko's Universal Approximation Theorem

## Theorem (Cybenko, 1989)

*Let  $\sigma$  be any continuous discriminatory function. Then  $S_\sigma$  is dense in  $C(K)$ : for any  $f \in C(K)$  and  $\epsilon > 0$ ,*

$$\exists G \in S_\sigma \text{ such that } \|f - G\|_\infty < \epsilon$$

The proof shifts the burden to showing specific activation functions are discriminatory.

## Proof of Cybenko's Theorem (Part 1)

1. Let  $S_\sigma$  be the subspace of network functions. Goal:  $\overline{S_\sigma} = C(K)$ .
2. **Assume contradiction:**  $\overline{S_\sigma} \neq C(K)$ .
3. By **Hahn-Banach**:  $\exists L \in C(K)^*, L \neq 0$ , with  $L(G) = 0$  for all  $G \in S_\sigma$ .
4. By **Riesz Representation**:

$$L(f) = \int_K f(x) d\mu(x), \quad \mu \neq 0$$

## Proof of Cybenko's Theorem (Part 2)

5. Since  $x \mapsto \sigma(w^T x + b) \in S_\sigma$ :

$$\int_K \sigma(w^T x + b) d\mu(x) = 0 \quad \forall w, b$$

6. By hypothesis,  $\sigma$  is **discriminatory**  $\implies \mu = 0$ .

7. **Contradiction** with  $\mu \neq 0$ .

8. Therefore  $\overline{S_\sigma} = C(K)$ . ■

# Proving Sigmoids are Discriminatory

## Lemma

*Any bounded, measurable sigmoidal function is discriminatory.*

### Proof sketch:

- Assume  $\int \sigma(w^T x + b) d\mu(x) = 0$  for all  $w, b$ .
- Construct  $h_\lambda(x) = \sigma(\lambda(w^T x + \theta) + \phi)$ .
- As  $\lambda \rightarrow \infty$ , pointwise limit  $\rightarrow$  indicator of half-space + boundary term.
- Dominated Convergence  $\rightarrow \mu(H^+) + \sigma(\phi)\mu(\Pi) = 0$ .
- Vary  $\phi$  (sigmoid is non-constant)  $\rightarrow \mu(\Pi) = 0, \mu(H^+) = 0$ .
- Half-spaces generate Borel  $\sigma$ -algebra  $\rightarrow \mu \equiv 0$ . (Dynkin's theorem) ■

# Leshno's Theorem: The Necessity of Non-Polynomials

**Theorem (Leshno, Lin, Pinkus, Schocken, 1993)**

$\sigma \in C(\mathbb{R})$  yields dense  $S_\sigma$  in  $C(K)$  for all compact  $K$  **if and only if**  $\sigma$  is **not a polynomial**.



# Leshno's Theorem: The Necessity of Non-Polynomials

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**Necessity** (easy): If  $\sigma$  is degree- $k$  polynomial,  $S_\sigma \subseteq \mathcal{P}_k$  (finite-dimensional  $\rightarrow$  closed  $\rightarrow$  not dense).

# Leshno's Theorem: Sufficiency Proof

**Sufficiency (harder):** If  $\sigma$  is not a polynomial:

- Translation + scaling  $\rightarrow$  derivatives via finite differences belong to  $\overline{S_\sigma}$
- $\frac{\partial^k}{\partial \lambda^k} \sigma(\lambda w^T x + b) \big|_{\lambda=0} = \sigma^{(k)}(b)(w^T x)^k \in \overline{S_\sigma}$
- Since  $\sigma$  is non-polynomial,  $\exists b$  with  $\sigma^{(k)}(b) \neq 0$  for all  $k$
- Choose appropriate  $w \rightarrow$  all monomials  $\rightarrow$  all polynomials
- **Stone-Weierstrass**  $\rightarrow \overline{S_\sigma} = C(K)$



## Theorem (Hornik's $L^p$ Approximation)

Let  $\mu$  be a finite Borel measure. If  $\sigma$  is bounded, continuous, and non-constant, then  $S_\sigma$  is dense in  $L^p(\mathbb{R}^n, \mu)$  for  $1 \leq p < \infty$ .

**Proof:**

$$f \in L^p \xrightarrow{\text{density of } C} g \in C(K) \xrightarrow{\text{Cybenko}} N \in S_\sigma \xrightarrow{\text{bound } \|\cdot\|_p \text{ by } \|\cdot\|_\infty} \|f - N\|_p < \epsilon$$



## Example: XOR with Heaviside

XOR:  $f(0,0) = 0$ ,  $f(1,0) = 1$ ,  $f(0,1) = 1$ ,  $f(1,1) = 0$ .

2 hidden neurons with Heaviside  $H$ :

- $h_1 = H(x_1 + x_2 - 0.5)$
- $h_2 = H(x_1 + x_2 - 1.5)$
- Output:  $y = h_1 - h_2$

| $(x_1, x_2)$ | $h_1$ | $h_2$ | $y$ |
|--------------|-------|-------|-----|
| (0,0)        | 0     | 0     | 0   |
| (1,0)        | 1     | 0     | 1   |
| (0,1)        | 1     | 0     | 1   |
| (1,1)        | 1     | 1     | 0   |

## Example: Polynomial Activation

$\sigma(x) = x^2$ . Show network cannot approximate  $f(x) = x^3$  on  $[-1, 1]$ .

Network with  $N$  neurons:

$$G(x) = \sum_{i=1}^N \alpha_i (w_i x + b_i)^2 = Ax^2 + Bx + C$$

**Always a quadratic**, no matter how large  $N$  is.

$\|x^3 - (Ax^2 + Bx + C)\|_\infty$  is always strictly positive (why?). Polynomial activations lack the universal property.

## Depth Efficiency & Separation

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# The Failure of Flat Representations

UAT says: 1 hidden layer is *sufficient*.

**But** at what cost?

- For many natural functions, shallow networks need **exponentially** many neurons
- The number of neurons scales as  $\exp(d)$  with input dimension

# The Failure of Flat Representations

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## Key Insight

Depth provides an **exponential gain** in representational efficiency.



# Piecewise Linear Complexity

For ReLU networks, complexity = number of linear regions.

## Theorem (Shallow Limit)

*A single-hidden-layer ReLU network with  $m$  neurons has at most  $m + 1$  linear segments.*

**Proof:**  $m$  kink points on  $\mathbb{R} \rightarrow$  at most  $m + 1$  contiguous intervals. On each interval, all ReLU states are fixed  $\rightarrow$  sum of linear functions = linear. ■

# The Multiplicative Power of Composition

## Theorem (Compositional Complexity)

If  $f, g$  are CPW functions with  $S_f, S_g$  segments, then  $f \circ g$  has at most  $S_f \cdot S_g$  segments.

## Insight

- **Addition** adds segments:  $m + m = 2m$
- **Composition** multiplies segments:  $m \cdot m = m^2$

## Definition (The Base Sawtooth)

$g : [0, 1] \rightarrow [0, 1]$ :

$$g(x) = \begin{cases} 2x & 0 \leq x < 1/2 \\ 2(1-x) & 1/2 \leq x \leq 1 \end{cases}$$

*Implemented by a 2-layer ReLU net with 3 units:*

$$g(x) = \sigma(2x) - 2\sigma(2x - 1) + \sigma(2x - 2)$$

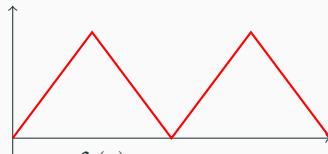
# Telgarsky's Sawtooth Construction (cont.)

## Definition (The Iterated Map)

$$f_L(x) = (g \circ g \circ \cdots \circ g)(x) \quad (L \text{ times})$$



$g(x)$ : 2 segments



$f_2(x)$ : 4 segments

# The Exponential Explosion

## Theorem

$f_L(x)$  has exactly  $2^L$  linear segments.

### Proof by Induction:

- **Base** ( $L = 1$ ):  $f_1 = g$  has 2 segments.
- **Inductive step:**  $f_{L-1}$  has  $2^{L-1}$  segments, each mapping  $[a, b] \rightarrow [0, 1]$ .
- Applying  $g$  splits each into 2  $\rightarrow$  total  $2 \cdot 2^{L-1} = 2^L$ . ■

# Telgarsky's Separation Theorem

## Theorem (Telgarsky, 2016)

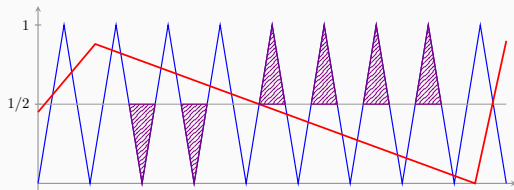
For any  $L \geq 3$ , there exists  $f_L$  computable by a ReLU network of depth  $2L$  and width 3 such that any shallow ReLU network with  $m \leq 2^{L-2}$  neurons satisfies:

$$\int_0^1 |f_L(x) - S(x)| dx \geq \frac{1}{32}$$

Shallow networks **cannot** track the oscillations without exponential width.

## Quick Proof

1. The constant line at  $\frac{1}{2}$  divide  $f_L$  into  $2^L - 1$  small triangles. Each small triangle has area  $2^{-L-2}$ . Proof by induction.
2.  $S(x)$  has at most  $2^{L-2} + 1$  linear components.
3. Track how  $S(x)$  passes through the line. When  $S(x)$  is above, we count the small triangle below; Otherwise, we count the small triangle above.



- 4 For each linear component, it can only pass the line once (generically at a point), so we do not count the triangle at that point. We also do not count the triangle at the right-end point (so no repeated counts), there are at most  $2 \cdot (2^{L-2} + 1)$  triangles not from above or below.
- 5 Only half of the triangles are counted,

$$\int_0^1 |f_L - S| dx \geq \frac{1}{2} (2^L - 1 - 2 \cdot (2^{L-2} + 1)) \cdot 2^{-L-2} = \frac{1}{16} - 3 \cdot 2^{-L-3} \geq \frac{1}{32}$$



# Multidimensional Separation

A 2-layer network computes **ridge functions**:

$$N(x) = \sum a_i \sigma(w_i^T x + b_i)$$

Each term is constant in  $d - 1$  directions (like a ridge)!

## Theorem (Eldan-Shamir, 2016)

*A radial function  $f(x) = \phi(\|x\|)$  can be  $\epsilon$ -approximated by a 3-layer network with  $\text{poly}(d)$  neurons, but requires  $\exp(d)$  neurons for any 2-layer network.*

The proof is too long to present here.

# Harmonic Perspectives

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# The Curse of Dimensionality

## Classical (computable) approximation

To achieve error  $\epsilon$  for a function with  $k$  continuous derivatives in  $d$  dimensions:

$$\text{Parameters needed} = O(\epsilon^{-d/k})$$

For high-dimensional functions ( $d \gg 1$ ), this means they are generally very difficult to approximate.

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For high-dimensional functions ( $d \gg 1$ ), this means they are generally very difficult to approximate.

## Question

Yet neural networks routinely solve high-dimensional problems reasonably. But **why?**

## Definition (Fourier Transform)

For  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ , define:

$$\hat{f}(\omega) = \int_{\mathbb{R}^d} f(x) e^{-i\omega \cdot x} dx$$

## Definition (Barron Norm)

For  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ , define:

$$C_f = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} \|\omega\|_2 |\hat{f}(\omega)| d\omega$$

Functions with  $C_f < \infty$  belong to **Barron Space** — they concentrate energy at low frequencies.

The Barron norm measures “spectral simplicity” — the first moment of Fourier magnitude.

## Theorem (Barron, 1993)

For any  $f$  with  $C_f < \infty$ , there exists a single-hidden-layer network  $f_n$  with  $n$  neurons such that:

$$\int_{\|x\| \leq r} |f(x) - f_n(x)|^2 d\mu(x) \leq \frac{(2rC_f)^2}{n}$$

Error depends on  $n$ , **not** on dimension  $d$ . The curse seems broken!

# Proof of Barron's Theorem (Part 1)

1. **Fourier representation** (assuming  $f(0) = 0$ ):

$$f(x) = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} (e^{i\omega^T x} - 1) \hat{f}(\omega) d\omega$$

2. **Probabilistic interpretation:** define probability measure  $\Lambda$  with  $d\Lambda \propto \|\omega\| |\hat{f}(\omega)| d\omega$ :

$$f(x) = C_f \mathbb{E}_{\omega \sim \Lambda} \left[ \frac{e^{i\omega^T x} - 1}{\|\omega\|} e^{i\theta(\omega)} \right]$$

## Proof of Barron's Theorem (Part 2)

3. **Maurey's Lemma:** If a function is in the **convex hull** of a bounded set  $G$  with  $\|g\| \leq B$ , then an average of  $n$  elements approximates it with error  $\leq B^2/n$ .
4. Ridge functions are bounded by  $r$  (since  $|e^{iz} - 1| \leq |z|$ ).
5. Sampling  $n$  frequencies  $\rightarrow$  constructing  $n$  neurons  $\rightarrow O(1/n)$  error.

$d$  does not appear explicitly. ■



## Example: Barron Norm of Step Function

$f(x) = H(x)$  (Heaviside step).

Fourier transform:  $\hat{f}(\omega) = 1/(i\omega)$ .

$$C_f = \frac{1}{2\pi} \int |\omega| \cdot \frac{1}{|\omega|} d\omega = \frac{1}{2\pi} \int 1 d\omega \rightarrow \infty$$

### Conclusion

Discontinuous step function is **not** in Barron space.  $\rightarrow$  Explains Gibbs phenomenon — networks struggle with sharp discontinuities.

# Attain Barron's Bound

The Barron's result establishes only the existence of a shallow neural network, without providing a computationally tractable algorithm.

Jones (1992) presented a greedy algorithm that iteratively constructs a shallow neural network in  $n$  steps as follows.

$$(\alpha', g') = \arg \min_{\alpha \in [0,1], g \in K(\sigma)} \|f - (\alpha h_{s-1} + (1 - \alpha)g)\|,$$

and generate  $h_s = \alpha' h_{s-1} + (1 - \alpha')g'$ . The algorithm can achieve the desired  $O(n^{-1/2})$  bound, but its search space is large (depends on the dimension).

## Exercise 1

Prove the algorithm can indeed attain  $O(n^{-1/2})$  and implement it.

# Curse of Dimensionality: Revisited

It seems Barron's result “defeats” the curse of dimensionality.

**Reality:** The Barron norm itself can grow exponentially in  $d$ .

## Example (Bessel Kernel)

$f(x) = (1 + |x|^2)^{-d/2}$ . Its Fourier transform is the Bessel kernel  $\hat{f}(\zeta) = \frac{2\pi^{d/2}}{\Gamma(d/2)} K_0(|\zeta|)$ .

The Barron norm:

$$\|f\|_{\mathcal{B}} = (2\pi)^{d/2} \left( \frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \right)^2 \sim e^{O(d)}$$

In high dimensions, the Barron space **excludes many smooth functions** — the probability of sampling a function with small Barron norm vanishes exponentially.

# Ridge Approximation Bounds

**Ridge functions:**  $\mathcal{R}_n = \{\sum_{i=1}^n k_i(w_i \cdot x) \mid k_i \in C(\mathbb{R}), w_i \in \mathbb{R}^d\}$ .

The key difference from a shallow network is that  $k_i$  are not fixed for ridge functions.

## Theorem (Maiorov, 1999 — Lower Bound)

For  $f \in W^{m,2}(B_1)$ ,  $\|f\|_{W^{m,2}} \leq 1$ :

$$\inf_{g \in \mathcal{R}_n} \|f - g\|_{L^2} \geq Cn^{-\frac{m}{d-1}}$$

## Theorem (Maiorov, 1999 — Upper Bound)

There exists  $\sigma \in C^\infty$ , sigmoidal, strictly increasing, such that:

$$\inf_{g \in \mathcal{S}_n} \|f - g\|_{L^2} \leq Cn^{-\frac{m}{d-1}}$$

where  $\mathcal{S}_n$  are shallow networks with activation  $\sigma$ .

# Ridge Approximation

The exponent  $-\frac{m}{d-1}$  (NOT  $-\frac{m}{d}$ ) is the optimal rate for ridge approximation. The optimal rate requires a very careful design for  $\sigma$ .

**Key idea for the upper bound:** any ridge function can be approximated by ONE “nice” activation function.

## Theorem

*There exists  $\sigma \in C^\infty$ , sigmoidal, strictly increasing, and for any  $g \in \mathcal{R}_n$  and  $\epsilon > 0$ , there are parameters  $(c_i, w_i, b_i)$  that*

$$\left| g(x) - \sum_{i=1}^{n+d+1} c_i \sigma(w_i \cdot x + b_i) \right| < \epsilon, \quad x \in X.$$

Rely on the **separability of continuous functions**.

1. Any continuous function on  $[-1, 1]$  can be approximated by polynomials.
2. Any polynomial coefficient can be approximated by rational numbers.
3. There is a countable set  $A$  of functions dense in continuous functions.
4. Construct  $\sigma$  to contain ALL of the functions in  $A$ .
5. In order to behave nicely, these functions need some linear basis to connect them.

### Question

How to find the approximation within that bound  $O(n^{-m/(d-1)})$ ?

This bound beats the **curse of dimensionality**.

# Stable vs. Unstable Approximation

## Definition (Nonlinear $n$ -Width)

$$h_n(K; \mathcal{X}) = \inf_{\mathcal{E}_n, \mathcal{D}_n} \sup_{f \in K} \|f - \mathcal{D}_n(\mathcal{E}_n(f))\|_{\mathcal{X}}$$

where  $\mathcal{E}_n : K \rightarrow \mathbb{R}^n$  is **continuous**,  $\mathcal{D}_n : \mathbb{R}^n \rightarrow K$  arbitrary.

## Theorem ( DeVore, 1989)

For the Sobolev ball  $W^{m,p}(B_1)$  in  $L^p$ :

$$h_n(W^{m,p}; L^p) \asymp n^{-m/d}$$

**Implication:** The optimal  $n^{-m/(d-1)}$  rate **cannot** be achieved by any continuous algorithm. Iterative training algorithms may NOT be continuous.

# Kolmogorov Superposition

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## Hilbert's 13th Problem

**Hilbert (1900):** Can every continuous function of 3 variables be expressed as a composition of continuous functions of 2 variables?

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**Kolmogorov (1957) and Arnold (1957):** Yes — and even stronger:

## Theorem

Any multivariate continuous function is a **finite superposition** of continuous functions of **one variable** and addition.

# Kolmogorov-Arnold Representation Theorem

## Theorem (Kolmogorov-Arnold, 1957)

*For any continuous  $f : [0, 1]^n \rightarrow \mathbb{R}$ , there exist  $2n + 1$  continuous  $\Phi_q : \mathbb{R} \rightarrow \mathbb{R}$  and  $n(2n + 1)$  continuous  $\psi_{q,p} : [0, 1] \rightarrow \mathbb{R}$  such that:*

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left( \sum_{p=1}^n \psi_{q,p}(x_p) \right)$$

**Key implication:** Multivariate complexity reduces to univariate complexity.

# KST: Kolmogorov Superposition Theorem

## Theorem (KST, Original — Kolmogorov, 1957)

For any continuous  $f : [0, 1]^d \rightarrow \mathbb{R}$ , there exist continuous  $\phi_{ij} : [0, 1] \rightarrow \mathbb{R}$ ,  $\Psi_j \in C[0, 1]$  ( $i \in [d]$ ,  $j \in [2d + 1]$ ) such that:

$$f(x_1, \dots, x_d) = \sum_{j=1}^{2d+1} \Psi_j \left( \sum_{i=1}^d \phi_{ij}(x_i) \right)$$

The inner functions  $\phi_{ij}$  are **universal** — independent of  $f$ .

## Theorem (KST, Variant)

The outer functions can be shared ( $\Psi_j = \Psi$  for all  $j$ ) with constants  $c_i$ , and  $2d + 1$  terms is **optimal**.

# From KST to Two-Hidden-Layer Networks

## Theorem (Maierov, 1999)

There exists  $\sigma \in C^\infty$ , strictly increasing, sigmoidal, such that for any  $f \in C[0, 1]^d$ :

$$\inf_{\theta} \sup_{x \in [0, 1]^d} \left| f(x) - \sum_{i=1}^{4d+3} w_i \sigma \left( \sum_{j=1}^{2d+1} c_{ij} \sigma(w_{ij} \cdot x + b_{ij}) + B_i \right) \right| = 0$$

**How:** Use the pathological  $\sigma$  to represent the KST inner/outer  $\Psi$  functions.  $2d + 1$  outer  $\Psi$ 's +  $2(d + 1)$  linear functions =  $4d + 3$  total.

**Caveat:** The approximation exists but CANNOT be found by a **continuous** (stable) algorithm.

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